

GLACIERS AND ICE SHEETS

Most of Earth’s glacial ice is concentrated in Antarctica and Greenland, but glaciers are found on every major landmass except Australia. Most exist either in the polar regions or in mountainous zones where the climate is cold enough, for at least a part of the year, for snow to

accumulate and ultimately become consolidated as ice. The amount of snowfall a glacier receives is crucial for its survival; regions that have suitably cold winter temperatures but little snowfall, such as Siberia, do not have the ice accumulation necessary for glacier formation.

Europe

The Kongsvegen Glacier

Location Svalbard archipelago, on the western side of Spitsbergen

One of more than 1,500 glaciers on Spitsbergen, Kongsvegen shares its lower end (terminus) in the seawater of Isfjorden with another glacier, Kronebreen. Both calve ice blocks into the waters of the fjord. Although Kongsvegen’s terminus is now slowly retreating, it is a surge glacier, meaning that periodically it advances rapidly. It last surged in 1948, pushing several miles along Isfjorden. It is a polythermal glacier, being warmer near the base than near the surface.

The Rhône Glacier

Location The Bernese Alps, Switzerland

At the head of an impressive U-shaped valley, which runs roughly northeast to southwest below the giant peak of Finsteraarhorn, the glacier is the source of the Rhône River (see p.372) and one of the main providers of water for Lake Geneva. It is about 5 miles (8km) long, although its snout has retreated 0.6 mile (1km) since 1880. Every summer, sections of the ice are covered with white blankets in an attempt to reduce ice melt.

The Pasterze Glacier

Location The eastern Alps, Austria

At just over 5 miles (8km) from head to terminus, Pasterze is the longest glacier in the eastern Alps. It is also located in one of the most

dramatic settings, with Austria’s highest mountain, Grossglockner, towering above it. The glacier’s head is Johannisberg, at 11,329 ft (3,453m), and at its snout it gives rise to the Moll River. Pasterze is currently retreating at a rate of 33 ft (10m) a year, and its volume has halved since the first measurements were taken in the mid-19th century.

Other glaciers and ice sheets in Europe

- The Aletsch Glacier » p.151
- The Jostedalsbreen Icefield » p.149
- Mer de Glace » p.150
- The Monaco Glacier » p.148
- Vatnajökull » pp.146–47



▲ ICE CAVE AND MELTWATER UNDERNEATH THE PASTERZE GLACIER